



KPMG IN INDIA

Oil & Gas scenario in India

INFRASTRUCTURE & GOVERNMENT

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Acronyms Used



E&P	Exploration & Production
MT	Metric Tonne
MMT	Million Metric Tonne
MMSCMD	Million Standard Cubic Meter Per Day
NELP	New Exploration Licensing Policy
SKO	Straight Run Kerosene
NG	Natural Gas
MoPNG	Ministry of Petroleum and Natural Gas

Executive Summary

The India Opportunity in Energy Sector



The Indian economy is a net importer of almost all forms of energy. This fact, coupled with the growing energy needs, has intensified discussions on energy security for the country. The government is actively seeking private participation in the energy chain and is also promoting acquisition of oil & gas reserves overseas. Back home, the rapidly growing Indian economy requires an investment of around USD 120 to 150 billion over the next five years in the energy sector. Strong private sector participation is required to complement public sector and bring in the required capabilities and technologies. Policies have increasingly recognized the need to promote private investment. Private interest in captive coal mining, oil & gas exploration and power sector has increased significantly and is also envisaged in nuclear sector, after the Indo-U.S. nuclear deal has been concluded.

There is also a shift towards market mechanisms with regulatory oversight in energy sector, especially in power and oil & gas. It is important that this transition is gradual till the supply side position improves. On a parallel level, the Government is making efforts to diversify the fuel basket by increasing shares of Natural Gas, Hydro and Nuclear energy. At the same time, both the Government and private sector companies are looking to acquire equity in energy assets abroad.

Energy transport infrastructure such as ports, railways, pipelines and power transmission networks need significant investment. The policy now allows private participation in all these areas and some private sector activity is already under way. Tariff reform in the energy sector and distribution reform in the power sector are two important steps that need to be carried out successfully. Tariff reform to phase out subsidies or to target them effectively and distribution reforms to bring efficiency in the power sector are vital.

The following sections highlight key opportunities in the different sectors:

Coal

India has vast reserves of coal, and participation of the private sector in captive mining, across different user industries, is an immediate opportunity. 38 coal mines¹ with reserves in excess of 6 billion tonnes have been identified and are in the process of being allotted, involving a total capital requirement of around USD 4-6 billion. Investment activity is also seen in other parts of value chain, including washeries.

¹Ministry of Coal and Mines



Oil

A number of private investors have entered this segment, attracted by the government's policies for upstream exploration and production. Six rounds of competitive bidding under the New Exploration Licensing Policy (NELP), have taken place and around 162 blocks were awarded and in-place volume estimated at 600 MMT² of oil & gas have been discovered. The seventh round of NELP has already been announced with a road show which commenced on January 8, 2008 in Mumbai. A total of 57 blocks have been offered. There is a new category of blocks called 'S' type for which no prior operatorship experience is required. In the downstream sector, there is a huge potential in refining, due to the strategic advantages of low cost and location; and India is already a net exporter of products. At present, the downstream marketing sector is also open to private participation.

Gas

Gas discoveries of around 700 bcm² in the last decade point towards tremendous potential. While in the near term, potential for LNG may be limited, due to inability of key sectors such as power to absorb high international prices, in the longer term it is likely there would be place for LNG as the share of Natural Gas in India's energy mix increases. On the demand side, an emerging area is auto-CNG and piped gas which together account for 7 percent of total gas demand in the last five years. In the next few years, at least 30 cities have been identified for city-wide gas coverage, by private and public sector players. The gas pipeline policy supports the development of a national gas grid, meant to create a common gas market across the country.

Nuclear

India has one of the largest reserves of the nuclear fuel - thorium. However, the nuclear energy programme will continue to be uranium-based, until commercial production based on thorium becomes feasible. If the Indo-U.S. nuclear deal goes through, the nuclear energy would receive a boost and private participation in this sector would be expected.

²Ministry of Petroleum and Natural Gas



Hydro

India is endowed with a hydroelectric potential of about 150,000 MW³. However, only 17 percent of the hydroelectric potential has been harnessed so far; with another 5 percent under various stages of development. Private participation in the hydro sector will be important to meet the target of an additional 45,000 MW of hydro capacity within the next 10 years. Various policy measures are being contemplated, to encourage private participation which seeks to address issues such as mitigating geological risks, resettlement and rehabilitation of project affected persons through Public Private Partnership (PPP) initiatives and incentives for performance. The revised hydro policy is currently under discussion by the Government of India.

Renewable Energy

India has a vast potential for renewable energy sources, especially in areas such as solar power, biomass and wind power. The current installed capacity of renewable energy is around 9220⁴ MW, constituting about 7.3 percent of India's total installed generation capacity. India is already the fourth largest in the world in terms of wind energy installations and we are seeing significant investment activity in this area. Technological breakthroughs could generate a quantum leap in the renewable energy sector, since India is well endowed with solar insolation.

Electricity

Generation

The government has envisaged a capacity addition of around 76,640 MW⁵ by 2012, with participation both from private and public sectors. Generation opportunities are encouraging on account of the opening of power trading, open access in transmission and distribution, and reforms in power sector. Recent developments in this area include the awarding of three Ultra Mega Power Projects (UMPPs) each 4000 MW in size, on the basis of competitive bidding. We are also beginning to see activity in relation to merchant power plants and the Government policy now encourages this, by assistance in providing fuel linkages and other clearances.

³Ministry of Power, Government of India

⁴Ministry of New and Renewable Energy, Government of India

⁵Blue Print for Power Development, Ministry of Power



Transmission

Private investment in the transmission sector can be done either independently or as a Joint Venture (JV). These participations are envisaged largely for creation of the National Grid⁶ along with the state-owned transmission utilities. The private sector participation is expected to be in projects requiring a capital outlay of around USD 4.5 billion⁷. In the last fiscal year, transmission projects in the western region covering approximately 1,500 km were awarded to the private sector on a Build, Own and Operate (BOO) basis in a competitive bidding process.

Distribution

Opportunities in distribution currently seem to be focusing on the franchising route. A number of distribution areas have been put up for private participation through the franchising route in recent months. While privatization of existing distribution utilities is possible, good opportunities are few, owing to the large risks involved. The political dispensation also does not seem to favor privatization of distribution at this time.

Trading

Power trading, as an activity, is evolving rapidly in India. Currently, around 15 GWh of electricity is traded every year and there are four or five large trading players. Merchant power plants, open access and the move to set up a Power Exchange, will all give a fillip to power trading.

Energy savings and Demand Side Management

A recent study estimated an immediate energy saving potential of 54,500 million kWh and peak saving of 9,240 MW. This has an investment potential of USD 3 billion. In spite of the above opportunities, a lot remains to be done, in terms of strengthening and building the regulatory institutions that will allow Government to distance itself from operational decision making and make the reform process more transparent and sustainable. For example, the electricity regulatory commissions would need more independence to deal with tariff design, market structure development, etc. and need to be given an environment to operate without political interference.

In the following chapters, starting with an overview of India's energy scenario, oil and gas sector has been discussed in greater details; highlighting key issues, the policy and regulatory framework and emerging areas for investments.

⁶Formation of the National Grid is a plan for strengthening of the inter-state and inter-regional transmission network that will enable unrestricted flow of electricity across regions and enable development of a deep electricity market

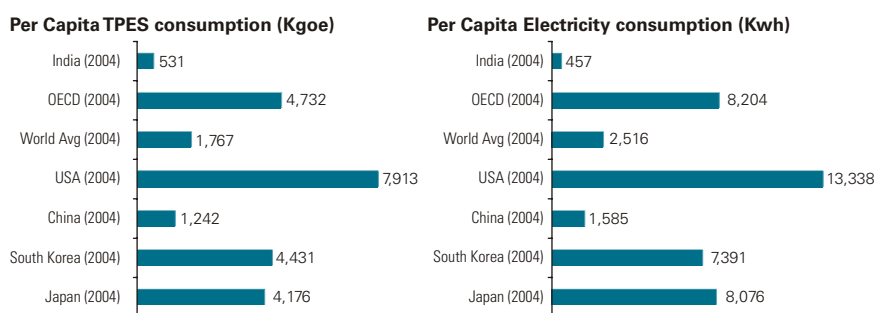
⁷Source: Ministry of Power, Government of India

Overview of India's Energy Position



By world standards, India's current level of energy consumption is very low. For the year 2004-05, the total annual energy consumption for India is estimated at 572 Mtoe (million tons oil equivalent) and the per capita consumption at 531 kgoe (kilograms oil equivalent). (Exhibit 1)

Exhibit 1: India's per capita energy consumption compared to other countries



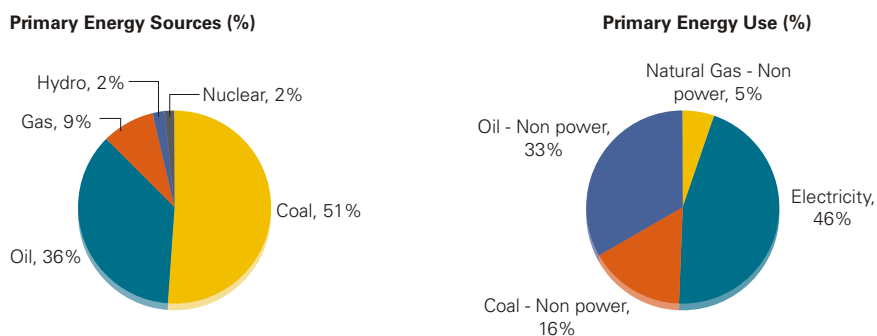
Source: International Energy Agency, KeyWorld Statistics 2006

With a target GDP growth rate of 7-8 percent and an estimated energy elasticity of 0.80, energy requirement is expected to grow at 5.6-6.4 percent. This would mean a four-fold increased in India's energy requirement over the next 25 years.

India's Current Energy Basket

While India is well-endowed in coal, 71 percent of its oil needs are met by imports. Exhibit 2 reflects only primary energy sources that are commercially exploited. Rural India is predominantly dependent on traditional fuel sources like firewood, animal dung and biomass, estimated at around 143 Mtoe per annum or approximately 44 percent of total primary energy use⁸.

Exhibit 2: India's composition of energy sources and usage



Source: Planning Commission of India, 2006

⁸Planning Commission of India



Future Energy Requirements and Supply Options

Given the present growth rate of 5 percent in coal production, India's extractable reserves would be exhausted in 45⁹ years, and hence there is a greater need to look at sustainable and cleaner fuels. Recent discoveries hold promise for India's gas reserves and coal bed methane. On the nuclear front, advanced technology needs to be infused before being put for commercial use. Renewable energy, especially wind and solar power is expected to grow rapidly and supplement the short term requirements. Over the longer term, it is expected to gain strategic importance as a sustainable fuel that would help build self-reliance in energy sources. The following figure details the estimated energy reserves in the country.

Exhibit 3: Estimated energy reserves

Resource	Unit	Reserves
Coal - Extractable	Mtoe	13,489
Oil	Mtoe	786
Gas – including coal bed methane	Mtoe	1,866
Uranium – metal	Tonnes	61,000
Thorium – metal	Tonnes	225,000
Hydel	MW	150,000

Source: Planning Commission of India, 2006

Different scenarios developed both on supply-side and demand-side are detailed as follows:

- **Energy efficiency in end-use:** Efficient energy use in industry, lighting, home appliances etc. could possibly lower the energy needs by 142 Mtoe in 2031-32 (7.5 percent of total requirement).
- **Increase in rail road share of freight:** Presently, most of freight traffic is carried by roads. If the share of railways in freight increases from the current 32 percent to 50 percent by 2031-32, there would be an estimated energy saving of 34 Mtoe in 2031-32 (1.8 percent of total requirement)
- **Increase in transportation efficiency:** Use of mass transport and better utilization and fuel efficiency of vehicles could save upto 81 Mtoe of energy by 2032 (4.3 percent of total requirement)
- **Efficiencies in thermal power generation:** Increase in thermal generation efficiency from the present 31 percent to 38-40 percent through use of super critical boiler technologies could lead to a savings of 111 Mtoe in 2031-32 (or 5.8 percent of the total energy requirement).

⁹Integrated Energy Policy, 2006



Together, there is a potential to save upto 351 Mtoe by 2032 (19 percent of total requirement).

On the supply side, the following options are envisaged:

- Fully exploiting India's hydro potential of 150,000 MW from current level of 32,326 MW
- Successful development of Fast Breeder Reactor (FBR) technology and Advance Heavy Water Reactor (AHWR) will scale up nuclear generation.
- Development of Natural Gas sources (indigenous, pipeline import or LNG) for power generation
- Development of renewable energy sources (solar power, bio-diesel and wind energy).

The range of utilization of different fuels in 2032, as compared to current levels is shown below:

Exhibit 4: Comparison of energy utilization in 2031-32 with present

Resource	Utilisation in 2031-32 (Mtoe)	Current utilisation (Mtoe)
Oil	350-486	119
Natural Gas (including CBM)	104-150	29
Coal	632-1022	167
Hydro	13-35	7
Nuclear	76-98	5
Solar	1200	<1
Wind	10	<1
Fuel wood	620	140
Ethanol	10	<1
Bio diesel	20	<1

Source: Planning Commission of India, 2006



Key Imperatives for India

To meet its large and growing energy needs, there are certain key imperatives for the Indian energy sector:

Private sector investment needs to complement public sector

Reliable and economic energy supply will require investment of capital as well as capabilities and efforts from both public and private sectors. The government is taking several steps to attract private players to this sector, which will need investments of around USD 120 to 150 billion over the next five years. Further clarity in areas including pricing of products and stability in policy framework is essential to encourage private investment.

Encourage market mechanisms with a credible and independent regulatory oversight

Market mechanisms will bring in efficiencies, and encourage investments by minimizing regulatory risks. With an improving supply-side situation, market mechanisms have been gradually introduced in the various segments of the energy chain, and this needs to be extended to other sectors like coal block allocation to encourage private sector participation.

Reduce vulnerability to price and supply shocks

The biggest challenge is to replace coal (expected to be exhaustible in 45 years), representing 51 percent of the energy basket, and oil which is heavily dependent on international supply in the short term towards Natural Gas, Hydro and renewable sources. Apart from diversifying the basket, enhancing domestic production and taking equity positions in energy resources abroad, are also necessary steps in reducing the effects of fuel price shocks.

Bringing in efficiency and enhancing capacity in energy transport infrastructure

To reduce the high inter-regional disparity to match demand and supply centres, significant investments in ports and railway, pipeline and storage network and infrastructure are underway. Enhanced transmission capacity will also enable the exploitation of Himalayan hydel resources. Private investments in ports, freight corridors, pipelines and power transmission are already seen.



Tariff reform and power sector reform

Heavily distorted power and energy prices have resulted in inefficient end-use and energy choices. Policy measures with sufficient political will are required to address these issues. Distribution reforms to cut down on network losses due to theft and pilferage are also necessary.

Provide Government Support for Energy Efficiency

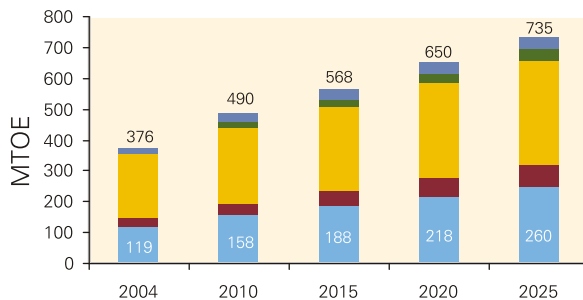
Policy framework incentivising energy efficiency is an urgent requirement. The environment should encourage energy efficiency companies to come up and operate profitably. In parallel, India is also emerging as an active market for Clean Development Mechanism (CDM) projects being conceptualized and registered with the Executive Board (EB). The growing awareness of the benefits from CDM would make this an important investment area and would give the necessary fillip for energy efficiency.

Oil Sector



Demand for oil, comprising of 36 percent of India's primary energy consumption, is expected to grow both in absolute and percentage terms to 196 MMT in 2011-12 and 250 MMT in 2024-25. To address the growing demand-supply gap, the government has stepped up exploration and production efforts through private participation under the NELP, and has also developed a more holistic strategy for acquisition of equity in oil abroad. ONGC Videsh Limited, the wholly owned subsidiary of Oil & Natural Gas Corporation Limited, is now active in 15 countries in Asia Pacific, Latin America, Middle East, Africa and Russia. Other Indian companies like Reliance Industries Limited (RIL), Gujarat State Petroleum Cor. Ltd. (GSPC) and Videocon are actively seeking oil & gas blocks across the globe.

Exhibit 5



Source BP and EIA:

■ Renewable ■ Nuclear ■ Coal ■ Gas ■ Oil

- Demand for oil is expected to grow from 119 MTOE from 2004 to 250 MTOE during 2025 at annual growth of 3.6 percent
- During the same period domestic production from existing developed reserves expected to grow at approximately 2.5 percent
- The gap in demand and output will catapult India to one of the largest consumers of crude oil along with China. The 2 countries will account for 35 percent of the world's incremental energy demand

Key issues facing the Sector

The following are the major issues faced by the Oil sector:

- **Energy security:** India imports more than 70 percent of its oil requirements. The dependence on import is expected to increase and could become as high as 90 percent in 2030. The government is following a two-prong approach. While the domestic oil & gas sector is now open to private participation through NELP bid rounds, Government is actively supporting acquisition of oil & gas blocks by Indian companies abroad
- **Absence of independent regulatory framework in the upstream industry:** The Indian Oil Sector is regulated by clauses on health, safety and environment, as also on certain critical aspects of operations and pricing, notably lacking structural reforms. The recently notified Petroleum Regulatory Board, is confined to the downstream industry, leaving the upstream industry under the general policy framework of the Directorate General of Hydrocarbons (DGH). Though, DGH has been given statutory credence through a recent amendment, it continues to be under the overall supervision of the Ministry of Petroleum and Natural Gas



- **Incidence of cross subsidy due to social obligations:** The incidence of the subsidy burden on LPG, SKO and diesel has now been spread out across to the private players as well. The differential pricing of Petrol and Diesel coupled with under-recovery on the sale of the two products, is adding to the market distortion. A long term approach for a market-based pricing mechanism is needed to boost investor confidence.

Policy and Regulatory Framework

- **Investment Policy:** There is a greater focus towards private investment. The government has allowed 100 percent FDI in exploration, creation of pipeline infrastructure, refining and in downstream retailing (subject to a minimum investment in mid-stream and upstream sector)
- **Pricing Policy:** Over the last five to six years, there has been a marked shift towards a market driven mechanism for prices. Several petroleum products have been de-controlled and allowed to be sold by private companies at market prices¹⁰. A government appointed committee has recommended wide-ranging measures including a shift from an 'import parity-based pricing' to a 'trade-based pricing', a reduction in custom duties on petrol and diesel, and the shifting of excise duty from an ad-valorem levy to a specific levy. However, prices of gasoline, diesel, LPG and Kerosene continue to be controlled by the Government.
- **Regulation:** The current upstream regulation is provided by Director General of Hydrocarbons (DGH) more on technical aspects than on pricing front. The midstream and downstream sectors have largely been unregulated. However, downstream regulation is now in place and operational, after the recent notification by MoPNG.
- **Tax structure:** India has a federal level tax structure governed by the provisions of the Income Tax Act, 1961. It has a wide network of tax treaties with over 80 countries across the globe to avoid double taxation of income. A brief overview of the India taxation system is appended as **Annexure A**. India also provides a customized tax regime for the upstream sector and non-resident service providers in relation to Exploration & Production operations. Further, the E&P players are entitled to a 100 percent tax holiday for seven years, with respect to profits earned from production of mineral oils. A brief overview of the regulatory and tax regime for upstream sector is appended in **Annexure B**.

¹⁰ Price of gasoline and diesel are still fixed by the Government, although the linkage to import parity price has strengthened significantly over the years



The India Opportunity

Investments under NELP

To increase upstream investments, the Ministry of Petroleum & Natural Gas (MoPNG) has introduced a transparent bidding process for allocation of oil & gas blocks. Six rounds of competitive bidding under the Government policy, named New Exploration Licensing Policy (NELP), have already been done, 162 blocks were awarded and in-place volume estimated at 600 MMT² of oil & gas have been discovered. The recent NELP-VI was a success with 165 bids being received from both domestic and international companies for exploration rights. Going forward, Directorate General of Hydrocarbons has already announced the seventh round of bidding for 57 blocks. Bids are due on April 11, 2008. The seventh round has a new category of blocks called "S" type for which no previous operatorship experience is needed. The seventh round further provides additional marks to foreign companies that are producing from deepwater areas. Some of the salient features of the latest NELP round are appended below:

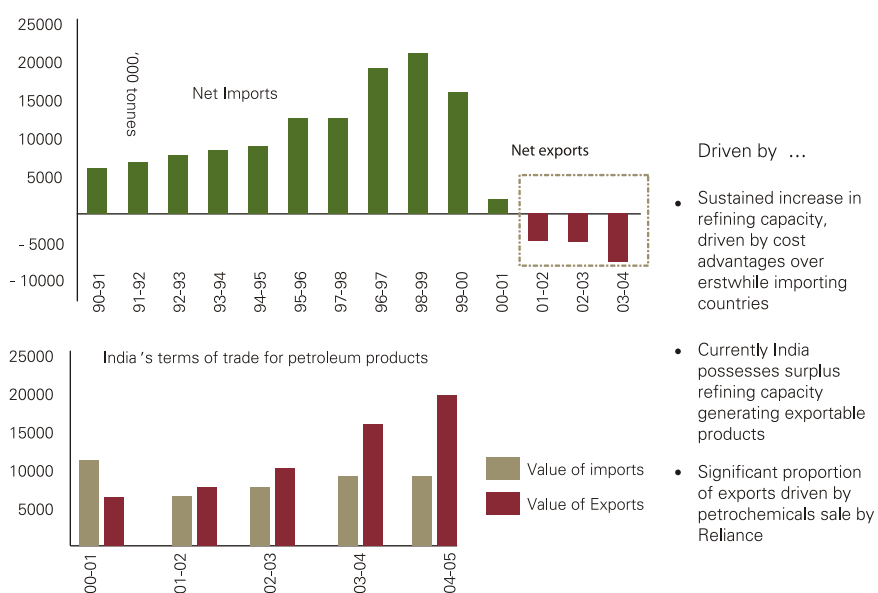
- Upto 100 percent participation by foreign companies
- Freedom to the contractor for marketing of oil & gas in the domestic market
- No carried interest by National Oil Companies
- Income tax holiday for seven years for the first seven years from the start of commercial production
- No custom duty on imports required for petroleum operations
- Option to amortize exploration and drilling expenditure over a period of 10 years from first commercial production.
- Royalty payable to the Government ranges between 5 percent to 12.50 percent, based on the type of operations



Destination India as refining hub

On the refining front, India enjoys significant advantages. It has lower construction and cash operating costs. India's strategic location en route of Middle East crude for East Asian and Pacific-rim markets is another key advantage. In fact, India possesses a surplus refining capacity and has already turned into a net exporter of products. The expected worldwide deficit caused by the shut down of small and uneconomical refineries around the world, bodes well for Indian refineries. The inability of oil majors to invest in refining assets during last decade and significant increase in the Chinese demand has accentuated the deficit. There is a planned addition of 80 - 90 MMTPA in India in the next four to five years to the existing capacity of around 149 MMTPA. Reliance Petroleum Limited is constructing a 28 MMTPA refinery at Jamnagar in Gujarat which is expected to become operational during the financial year 2008-09. Hindustan Petroleum Corporation Limited (HPCL) has entered into a partnership with Mittal Investments, to set up a 9 MMTPA refinery-cum-petrochemical complex at Bhatinda in Punjab.

Exhibit 6





Increased investment in fuel quality upgradations

Significant investments are planned for upgrading existing refineries to meet the stringent fuel specifications as per the domestic “Auto Fuel Policy” which mandates Euro IV norms by 2010 in the 11 designated cities and Euro III norms in rest of the country. More complex configurations are required to hedge against variation in crude supplies and to achieve cost competitiveness, by accommodating cheaper quality crude. The state-owned IOCL has planned investments of INR 230 billion for several refineries. HPCL and BPCL have planned similar investments for their Vishakhapatnam and Mumbai refineries respectively.

Building a strategic petroleum reserve through PPP

The Government has taken steps to set up strategic crude oil storage reserves at various locations, improving the country's ability to respond to short term supply disruptions. It is also exploring the possibility of increasing oil stockpiles through innovative schemes including the leasing of space to international traders, and building additional storage terminals through concessions.

Acquisition of overseas oil assets

MoPNG has conceived a more coordinated approach towards acquisition of overseas oil assets through joint forays. In addition, it is entering into bilateral and multilateral co-operation engagements with other countries in the areas of technology transfers, R&D, safety and training. Recently, India signed a MoU with China for joint bidding of hydrocarbon blocks.

Competition in the downstream (retail and institutional) segment

As per the Petroleum and Natural Gas Regulatory Board (PNGRB) Act, upcoming transmission network would have open access. Anticipating competition in the retail segment, major oil firms are expanding their retail network and forging alliances to offer non-fuel products and services as part of their overall proposition. On the institutional segment, incumbents are focusing on profitable segments, including ATF as well as specialty products like Hexane.

Gas Sector



Per capita consumption of Natural Gas in India is currently amongst the lowest in the world; at 29 cu m as compared with a world average of around 538 cu m¹¹. The present share of natural gas in the energy basket is only around 9 percent in India, compared to a world average of around 24 percent. However, demand for NG (at about 180 mmscmd) in the country has far outstripped supply (about 95 mmscmd), and there has been an increasing trend towards emergence of new NG demand as well as conversion from existing fuels to NG¹². Though India is a new entrant to natural gas (NG), the significance of the fuel can be gauged from the fact that, by 2025, the country is expected to rival both China and Japan in having the largest NG demand in Asia. Demand in each of these countries is expected to be in the range of 350 MMSCMD¹³. More than 50 percent of NG volume in the country is expected to be as cleaner and cheaper substitutes to petroleum products, with the rest as cleaner substitutes to coal in the power sector. In total, the share of NG in the fuel mix is expected to go up to 22 percent in 2031-32.

Key Issues

In contrast to oil, 80 percent of Indian NG demand is met from domestic sources. The key issues faced by this factor are as follows:

- **Domestic reserves/ production will not be sufficient:** Declining trends point to reduction in production to less than 50 percent of current production¹⁴ levels from the existing fields by 2015. Though, the new gas discoveries can contribute to about 100 MMSCMD of gas production. Even with this, the supply shall fall short of the projected demand. There is a renewed focus on identification of new supply sources through the NELP programs to attract greater investment in this area. Unattractive market prices in the last decade may also have contributed to a lag in production
- **Cross-border gas pipelines facing uncertainty, but attracting interest:** Unfavourable political environment and international climate have delayed cross-border pipeline projects. However, increasing LNG costs and the global history of high reliability of supply in transnational pipelines, are factors that are in favour of these projects
- **Inability to take international prices:** Seventy percent of the NG market is made up of the power and fertilizer sectors, who were not able to absorb international prices and were subject to an administered price mechanism, till a couple of years ago. Shortages in production have since doubled the prices, and shifted fuel preferences from gas to coal in the power sector. However, gas is still preferred to Naphtha, FO and LPG because of its competitive prices.

¹¹ Draft Report on the Expert Committee on Integrated Energy Policy, 2005

¹² GAIL. Infraline Natural Gas in India 2005

¹³ Energy Outlook 2004, HV 2025

¹⁴ Referred from LTGP statistics of DGH



Policy and Regulatory Framework

Over the past six years, the trend in natural gas regulation has been towards opening up the sector for greater investment, setting up an independent regulator to monitor post production activities, and enabling a transition from the administered control regime to a market driven mechanism. Significant regulatory bodies and issues in this sector include:

- **Petroleum & Natural Gas Regulatory Board Act, 2006 (PNGRB Act, 2006):** The authority to regulate all non-production activities in the petroleum and natural gas value chain is intended to promote consumer interests, reliability of supply and competition.
- **Policy for Development of Natural Gas Pipelines and City or Local Natural Gas Distribution Networks:** Through investments in critical infrastructure, this policy seeks to facilitate open access for all players without discrimination and promote competition. It also seeks to set up a Gas Advisory Board for the development of the pipeline network.
- **Gas Linkage Committee:** This committee seeks to manage gas allocation to eligible consumers. Though earlier linked with the Administered Price Mechanism (APM), new fields under the NELP are exempt from purview of GLC and can trade at market prices.
- **Infrastructure Status for gas pipelines:** The grant of 'Infrastructure Status' to the pipeline and storage sector translates into a number of benefits for companies engaged in transmission pipelines. However, if part of these benefits, are passed on to consumers, it could also lead to lower tariffs.
- **Foreign Direct Investment in NG Sector:** 100 percent FDI is allowed in exploration, pipeline infrastructure, LNG and trading segments. This is likely to bring forth significant investments. An integrated LNG policy is likely to be in place soon.



The India Opportunity

Domestic exploration of NG

The government sees significant domestic exploration potential, to match demand and supply. On an average, in-place volume of more than 70 bcm have been discovered annually. NELP provides significant benefits to private players in terms of 100 percent FDI, a seven year tax holiday, free marketing rights etc. NELP VI has already proved to be very successful with these initiatives. NELP VII has also opened up a new set of opportunities for investors. Significant finds are crucial to bridge the supply gap in domestic gas. After the formation of the National Gas Hydrate Program, gas hydrate exploration has also received considerable impetus. India is the third country to engage a specially designed vessel to carry out drilling activities in Indian waters for collecting cores for studies of gas hydrates habitation, contents, etc.

Is LNG the answer?

LNG as a substitute for costlier Naphtha has seen increasing absorption in the domestic market. However, due to a number of reasons, LNG would find it difficult to compete with other options in the short to medium term for power generation. The lack of a cross-country gas pipeline to enable transmission, the emphasis on coal as the preferred fuel for Ultra Mega Power Plants and the gradual emergence of Coal Bed Methane have made LNG increasingly uncompetitive for power generation. In addition, issues related to pricing and the limited potential of LNG supply sources to India need to be sorted out. However, in the long term, LNG is likely to be one of the most significant areas of investment.

Coal Bed Methane (CBM) and Underground Coal Gasification Opportunities

With proven reserves of 765 Mtoe and indicated reserves of between 1,260 and 2,340 Mtoe¹⁵, CBM could be another opportunity. CBM exploration has already been taken up seriously, with more than 26 blocks awarded so far and more to be taken up as part of future phases of CBM bidding. Compression of CBM and marketing as CNG could be exploited in potential industries as a monetization option for stranded gas.. A related exciting technology is that of underground coal gasification, which is already being exploited in Russia at a small level. Given India's large coal reserve, the UGC technology could potentially produce volumes of multiples of India's free natural gas reserve.

¹⁵ Draft Report of expert committee on Integrated Energy Policy



Emergence of the retail gas user

Gas is becoming a preferred fuel for retail user segment as a cheaper and cleaner fuel for domestic and transportation purposes. The growth of Auto CNG and Piped domestic gas in major Indian cities has sparked off a new demand spurt for NG. The fast pace of growth can be assessed from the fact that in the next few years, at least 30 cities would be embraced for city-wide gas coverage by private and public players, supported by regulation, as compared to the six cities today.

Development of a common gas market through National Gas Grid

The growth in each of the end user industries as well as the widespread growth of the retail segment would need to be supported by the appropriate infrastructure. The planned National Gas Grid connectivity is with a view to harmonize the operations and provide interconnectivity to different gas pipelines.

The New Gas Pipeline Policy announced by the Government provides a framework for development of a National Gas Grid. In addition to this, with the setting up of the Petroleum and Natural Gas Regulatory Board private interest is expected to increase tremendously in the pipeline infrastructure segment

Exhibit 7



Annexure A



Overview of the Indian Taxation System

Income Tax

India has a federal level tax structure governed by the provisions of the Income Tax Act, 1961.

Scope of total income

- A resident in India is liable to tax on its world wide income irrespective of the source of income
- A non resident in India is liable to tax on income received or deemed to be received in India or any income accruing or arising or deemed to be accruing or arising in India.

Scheme of Taxation

- Taxation of a person depends upon its legal status (a person being an individual, firm, company, etc.) and residential status
- Indian tax system recognizes an entity level taxation.

Other Features

- Loss carry forward permitted upto eight years, however, depreciation can be carried forward indefinitely
- No tax on remittance of profits by foreign companies (project office/branch office to head office)

Rates applicable for the financial year 2007 - 2008 are as follows:

Resource	Indian Company	Foreign Company
Corporate tax rate	33.99% *	42.23% *
Minimum Alternate tax	11.33% *	10.5575% *
Dividend Distribution tax	16.995%	N. A.
Fringe Benefit tax	33.99%	31.6725%
*In case net income exceeds INR 10 million		

Minimum Alternate Tax (MAT)

- MAT is applicable to a company, if tax payable by the company on its total income, as computed under the normal provisions, is less than 10 percent of its book profits
- Due to the MAT regime, a company may be required to pay tax even during tax holiday period
- In computing 'book profits' for MAT purposes, certain positive and negative adjustments are made to the net profit as shown in the books of account
- Carry forward and set off of MAT is available for seven subsequent years
- Set off is allowed to the extent of difference between tax on total income under normal provisions and MAT payable.

Dividend Distribution Tax (DDT)

- DDT is levied at the rate of 16.995 percent on the amount of dividend declared, distributed or paid by an Indian company
- Dividend from domestic companies is exempt from tax in the hands of recipient
- DDT is payable in addition to regular corporate income tax.

Fringe Benefit Tax (FBT)

- FBT is payable by an employer on the benefits provided or deemed to have been provided to the employees
- Tax is payable on value of fringe benefit as prescribed i.e. 5 percent, 20 percent or 100 percent of the costs incurred on such benefits.

Transfer Pricing provisions

- Transfer pricing provisions were introduced in the financial year 2001-02. Under these provisions, international transactions between associated enterprises are required to be computed with regard to their arm's length price
- The domestic law prescribes the information and documents which are required to be maintained by every person who has entered into an international transaction with its associated enterprises.

Taxation of Individuals

- Taxability of an individual is dependent on his/her residential status
- The residential status of an individual is determined on the basis of his/her physical presence in India
- Based on the satisfaction of certain conditions, an individual could be:
 - Resident and ordinarily resident (ROR)
 - Resident but not ordinarily resident (RNOR)
 - Non-resident (NR).

Taxability				
Residential Status	Worldwide Income		Indian Income	
	Received in India	Received outside India	Received in India	Received outside India
POR	✓	✓	✓	✓
RNOR*	✓	✗	✓	✓
NR	✓	✗	✓	✓

* Income derived by a RNOR from a business controlled or profession set up in India shall be taxable in India.

Tax rates applicable for the financial year 2007-2008 are

Taxable Income	Rate percent
Upto INR 110,000*	Nil
INR 110,000 - INR 150,000	10%
INR 150,000 - INR 250,000	20%
Above INR 250,000	30%

* Basic exemption limits for a resident woman is INR 145,000 and for a resident citizen (having age of 65 years) is INR 195,000

Note: The above tax rate would be further increased by surcharge of 10 percent if taxable income of the individual exceeds INR 10,00,000. Additionally, education cess of 3 percent would also be levied.

Key Indirect Taxes

Service Tax

- Service tax is applicable on identified services provided or received in India
- Current scope of taxable services is very wide and covers a vast majority of service categories
 - Engineering, management, scientific and technical consultancy, broadcasting, construction, IPR, insurance, manpower, communication, online access, training, cargo handling, business auxiliary services are some of the key categories
- Service tax is applicable at 12.36 percent
- Export of services are exempt from service tax - export determined as per prescribed rules
- Import of service also liable to service tax - import determined as per prescribed rules.

VAT Legislation

- Since its inception in April 2005, VAT has been implemented in all States and Union Territories with exception of Andaman and Nicobar and Lakshadweep
 - VAT is a multi-point taxation system entailing a VAT at every point of sale/lease
 - Dealers are allowed to avail credit of input tax on input and capital goods for set-off against out-put VAT
 - Common rate of tax adopted across all States with rates of 12.5 percent, 4 percent and 1 percent prescribed for different categories of goods. Also, some category of goods have been declared exempt from levy of VAT
 - Interstate sale of goods is not governed by VAT (liable to a central sales tax).

Custom Duty

- Custom Duty is payable on import of goods/equipments into India
- It is levied as per rates specified in the Customs Tariff Act
- Peak rate of Customs Duty is 10 percent.

Annexure B



Regulatory and Tax Regime for Upstream Sector

Regulatory

- FDI upto 100 percent is permitted under the automatic route in the upstream sector
- A foreign company can setup a project office or an Indian company for undertaking upstream operations in India.

Income Tax

There is a special mechanism for taxation of income of companies which have entered into a Production Sharing Contract (PSC) with the Government of India for undertaking exploration and production activities.

- As per these provisions, taxable profits of a tax payer, who has entered into a PSC with the Government for participation in the business of prospecting, exploration or production of mineral oil, to be determined in accordance with the special provisions contained in the PSC
- The provisions of the domestic tax law are deemed to be modified to that extent.

Special provision

- Specific allowances [in addition or in lieu of allowances under normal provisions] as specified in the PSC are permitted.
- The specific allowances relate to:
 - Expenditure by way of infructuous or abortive exploration
 - Expenditure incurred for exploration or drilling activities or services or assets used for these activities.

PSC

- Allowability of expenditure
 - Special deduction - 100 percent of exploration and drilling expenses (both capital and revenue allowed)
 - Other expenses (including production expenditure) allowed under normal provisions.
- Manner of deduction
 - Allowable expenditure is aggregated till the commencement of commercial production
 - Accumulated expenditure allowed in the year of commencement of commercial production or permitted to be amortized over a period of 10 years.

No Ring Fencing of Expenditure

- All unsuccessful exploration costs in other contract areas can be set off against income in the contract area in which commercial production has commenced.

Tax Holiday

- 100 percent tax holiday available in respect of profits earned from production of mineral oils (which includes petroleum and natural gas)
- Deduction is available for seven consecutive years from the time of commencement of commercial production
- However, companies availing deduction under these provisions would still be liable to pay MAT on 'book profits'.

Deductibility of Site Restoration Expenses

- Special deduction is available for site restoration expenses
- Amount of deduction being lower of:
 - Sum deposited either in a special account or in a "Site Restoration Account" or
 - 20 percent of the profits calculated in the prescribed manner.

Taxation of service providers

- **Applicability**
Special tax regime for non-resident service providers engaged in the business of providing services or facilities or supplying plant and machinery on hire in connection with prospecting for, or extraction or production of, mineral oils.
- **Mechanics**
10 percent of the gross receipts deemed to be business income resulting in an effective tax rate of 4.223 percent of gross revenues (rate as applicable for financial year 2007-08).
- **Option to claim lower profits, subject to following conditions:**
 - Keep/maintain books/documents
 - Get accounts tax audited
 - Furnish tax audit report
 - Compulsory scrutiny assessment

Custom Duty

Subject to certain procedures and conditions, Custom Duty exemption is available for:

- Equipments etc. imported for exclusive use in petroleum operations
- Specified goods required in connection with petroleum operations under specific exemption notification
- Parts and raw materials for manufacture of goods for the purpose of off-shore petroleum operations undertaken under specified contracts.

Service Tax

Relevant Service Tax Category

- Survey and exploration of mineral, oil & gas services (effective from 10 September 2004)
 - Includes geological, geophysical or other prospecting, surface and subsurface surveying or map making service, in relation to location or exploration of deposits of mineral, oil or gas
- Site formation and clearance services (effective from 16 June 2005)
 - Includes drilling, boring and core extraction services in relation to site formation and clearance, excavation and earth moving and demolition
- Mining services (effective from 1 June 2007)
 - Recently introduced to tax 'any service provided in relation to mining of minerals, oil & gas'
- Commercial or industrial construction
 - Includes construction of well head and civil works at site.
- Service tax also leviable on the following services:
 - Dredging services
 - Technical testing and analysis
 - Pipeline transportation
 - Cleaning (including services for tank, reservoir of commercial or industrial building and premise)

About KPMG in India

KPMG is the global network of professional services firms of KPMG International. Our member firms audit, tax and advisory services through industry focused, talented professionals who deliver value for the benefit of their clients and communities. With nearly 113,000 people worldwide, KPMG member firms provide services in 148 countries.

The member firms of KPMG International in India were established in September 1993. As members of a cohesive business unit, they respond to a client service environment by leveraging the resources of a global network of firms, providing detailed knowledge of local laws, regulations, markets and competition. We provide services to over 2,000 international and national clients, in India. KPMG has offices in India in Mumbai, Delhi, Bangalore, Chennai, Hyderabad, Kolkata and Pune. The firms in India have access to more than 2000 Indian and expatriate professionals, many of whom are internationally trained. We strive to provide rapid, performance-based, industry-focused and technology-enable services, which reflect a shared knowledge of global and local industries and our experience of the Indian business environment.

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